

Basics and Principles

Glossary

1. Core HCI Concepts

1. Human-Centered Design (HCD)

- A design philosophy that prioritizes user needs through research, prototyping, and iterative testing.
- *Key Principles (Gould & Lewis, 1985):*
 - **Early focus on users/tasks:** Understand user demographics and workflows.
 - **Empirical measurement:** Test prototypes with real users.
 - **Iterative design:** Refine based on feedback.

2. Mental Model

- A user's internal representation of how a system works, shaped by experience and beliefs.
- *Types:*
 - **Functional:** Knows *what* actions to take (e.g., clicking "save").
 - **Structural:** Understands *why* actions work (e.g., files save to cloud).
- *Example:* Thermostat users setting to "max" believing it heats faster (incorrect model).

3. Conceptual Model

- The designer's intended explanation of system functionality, communicated via UI elements (e.g., trash bin icon = deletion).

4. System Image (*Don Norman*)

- How a product's design (buttons, labels, documentation) conveys its operation to users.

5. Metaphor

- Using familiar real-world concepts to explain digital interfaces (e.g.,

"desktop" for files).

- *Limitation*: Breaks down if mismatched (e.g., "filing cabinets" ≠ cloud storage).

2. Design Processes & Methods

6. Iterative Design

- Cyclic process: Design → Test → Refine → Repeat.

7. Task Frequency Analysis

- Prioritizing UI elements based on how often tasks are performed (e.g., bold text > "all caps").

8. Hierarchical Task Analysis (HTA)

- Decomposing tasks into sub-goals (e.g., "Post photo" → "Open app" → "Upload image").

9. Persona

- Fictional archetypes representing key user groups (e.g., "Tech-Savvy Teen" for a social app).
- *Purpose*: Avoid designing for an "average" (nonexistent) user.

10. Usage Profile

- How different user types (novices, experts) interact with a system.

3. Key Principles & Pitfalls

11. "Make Mistakes Early"

- Identify usability issues in prototypes to reduce costly fixes later.

12. Gulf of Execution (*Don Norman*)

- The gap between user goals and system actions (e.g., unclear navigation).

13. Feature Bloat

- Adding unnecessary functions that complicate the UI.

14. Niche Targeting vs. Mass Appeal

- Designing for specific user groups (e.g., professionals) vs. broad audiences.
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4. Examples & Models

15. **Desktop Metaphor** (*Alan Kay*)
 - UI mimicking physical desktops (files, folders, trash).
 16. **Central Heating Mental Model**
 - *Correct*: Set thermostat to desired temp (structural understanding).
 - *Incorrect*: Set to "max" first (flawed functional model).
 17. **Designer-User Model Gap**
 - Mismatch between how designers envision a system and how users perceive it.
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5. Additional Terms

18. **Front-End Analysis**
 - Early-phase research to identify user needs/constraints.
 19. **Performance Support**
 - Tools aiding real-time tasks (e.g., in-app tutorials).
 20. **Organizational Structures**
 - How team workflows impact design (e.g., agile vs. waterfall).
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References

- **Norman, D. (2013):** *The Design of Everyday Things* (System Image, Gulf of Execution).
 - **Gould & Lewis (1985):** Usability principles.
 - **Cooper (2007):** Personas in design.
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